

MERAKER-EGGE, Norway

WMO: 012930

Lat: 63.411N

Lon: 11.727E

Elev: 171

StdP: 99.29

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.0	-17.7	-23.5	0.5	-20.8	-20.4	0.6	-17.1	10.3	0.6	9.2	-0.1	1.7	190	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	26.6	16.7	24.4	15.8	22.3	14.8	17.8	24.5	16.7	22.5	15.6	21.0	2.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	11.0	20.5	14.1	10.2	18.9	13.1	9.6	17.6	50.8	24.3	47.2	22.6	44.1	21.1	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	6.8	5.8	-23.2	29.6	2.8	2.3	-25.2	31.3	-26.8	32.6	-28.4	33.9	-30.5	35.6
			-23.4	19.8	2.8	1.4	-25.4	20.8	-27.0	21.6	-28.6	22.4	-30.6	23.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.7	-4.0	-3.2	-1.1	3.5	8.1	11.8	15.1	13.7	10.1	4.7	-0.1	-2.7
	DBStd	8.02	6.03	5.80	4.96	3.17	3.93	3.45	3.47	2.93	3.06	3.84	5.25	6.07
	HDD10.0	2354	433	371	343	195	87	19	1	3	35	169	304	394
	HDD18.3	4990	691	604	601	444	318	199	111	146	246	423	554	652
	CDD10.0	422	0	0	0	1	27	72	160	118	39	5	0	0
	CDD18.3	15	0	0	0	0	0	1	12	2	0	0	0	0
	CDH23.3	302	0	0	0	0	19	48	192	42	1	0	0	0
	CDH26.7	58	0	0	0	0	2	7	46	3	0	0	0	0
Wind	WSAvg	2.4	2.7	2.7	2.6	2.4	2.4	2.3	2.1	2.0	2.2	2.3	2.3	2.5
Precipitation	PrecAvg	1019	86	67	72	51	64	88	114	96	122	94	79	96
	PrecMax	1225	241	123	239	117	123	150	193	160	232	140	192	222
	PrecMin	814	1	2	12	14	11	27	61	36	37	43	0	33
	PrecStd	115	55	35	55	27	24	35	36	36	49	26	52	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	8.2	10.9	17.4	26.2	27.9	30.9	27.1	22.3	17.1	12.4	8.5
		MCWB	4.5	4.4	5.2	9.7	15.2	16.0	19.2	16.8	14.2	11.8	8.9	5.5
	2%	DB	5.7	6.3	8.0	14.5	22.4	24.3	27.5	24.3	20.0	14.3	10.0	7.2
		MCWB	3.3	3.3	4.3	7.8	13.2	14.8	17.6	16.0	14.1	9.8	7.2	4.5
	5%	DB	4.6	5.1	6.6	12.0	19.3	21.7	25.4	22.3	18.0	12.7	8.4	5.9
		MCWB	2.6	2.6	3.7	6.3	12.2	14.0	16.6	15.3	13.0	9.2	5.8	3.5
	10%	DB	3.3	3.9	5.2	9.5	16.0	19.3	23.1	20.2	15.9	10.8	6.6	4.7
		MCWB	1.6	1.8	2.5	5.2	10.2	12.9	15.8	14.5	11.7	8.0	4.2	2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	5.0	6.3	10.1	15.7	17.2	20.4	18.3	15.7	12.9	9.3	6.1
		MCDB	6.5	7.1	8.5	16.8	24.4	24.4	28.7	24.3	20.4	14.8	11.7	7.4
	2%	WB	3.9	3.9	5.1	8.3	13.9	15.8	18.8	17.0	14.5	10.4	7.6	4.9
		MCDB	5.2	5.8	7.6	13.6	20.8	22.7	26.3	22.5	19.0	13.6	9.6	6.8
	5%	WB	2.8	3.0	4.0	6.8	12.3	14.6	17.4	16.0	13.3	9.3	6.0	3.8
		MCDB	4.3	4.4	6.3	11.1	18.5	20.4	23.7	20.8	17.0	12.3	8.1	5.5
	10%	WB	1.6	1.9	2.8	5.6	10.6	13.3	16.2	15.0	12.1	8.2	4.4	2.6
		MCDB	3.0	3.8	4.9	9.0	15.8	18.4	21.9	19.3	15.5	10.8	6.3	4.4
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	8.7	9.6	10.2	9.7	10.7	10.5	9.3	7.4	6.1	6.3
		MCDBR	5.7	5.9	9.2	14.3	16.6	15.4	16.3	14.6	13.2	9.8	8.2	7.0
		MCWBR	4.2	4.2	6.1	8.5	8.7	7.6	7.9	7.7	8.0	6.4	5.9	5.5
	5% WB	MCDBR	5.4	5.8	8.1	12.1	15.5	13.6	14.9	12.5	12.4	8.7	8.0	6.9
		MCWBR	4.3	4.5	5.6	7.5	8.4	7.0	7.6	7.0	7.7	5.9	6.0	5.7
Clear-Sky Solar Irradiance	taub		0.219	0.280	0.313	0.342	0.351	0.349	0.358	0.346	0.321	0.300	0.230	N/A
	taud		2.007	2.179	2.201	2.257	2.377	2.450	2.444	2.473	2.529	2.418	2.083	N/A
	Ebn at Noon		390	647	774	834	862	872	851	831	784	639	371	N/A
	Edh at Noon		30	61	90	106	103	99	97	86	67	49	27	N/A
All-Sky Solar Radiation	RadAvg		0.16	0.69	1.98	3.78	4.58	4.79	4.64	3.55	2.13	0.94	0.25	0.08
	RadStd		0.02	0.10	0.27	0.30	0.31	0.47	0.49	0.40	0.21	0.12	0.04	0.01

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-142	-178	N/A	N/A

Nomenclature: See separate page